

ANTIANGINAL DRUGS

Introductory case

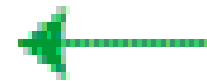
- A 77 year old professor, has been taking sublingual nitroglycerine PRN for exercise-related chest pain (angina) for the past 2 years. Recently metoprolol was also added to his daily drug regimen. On a follow-up visit 3 weeks later, he states that his need for nitroglycerin has been reduced since he started taking the beta blocker.
 - What is the most common pathophysiological mechanism underlying exercise induced angina?
 - How does nitroglycerine alleviate his angina?
 - Why is nitroglycerine taken sublingually?
 - How does metoprolol reduce angina?
 - Why did metoprolol reduce his need for taking nitroglycerin?

- **Angina pectoris**
 - sudden, severe, pressing chest pain radiating to the neck, jaw, back, and arms
 - When Oxygen supply to the heart is insufficient to meet oxygen demand
- **The imbalance between oxygen delivery and utilization may result from**
 - exertion
 - a spasm of the vascular smooth muscle, or
 - obstruction of blood vessels caused by atherosclerotic lesions

Decreased
coronary
blood flow



Angina
(Chest Pain)



Increased
oxygen
consumption



Vasospasm
Fixed Stenosis
Thrombus



Increased:
Heart Rate
Contractility
Afterload
Preload

Ischemic heart disease

Coronary
thrombosis

Myocardial infarction

Transient
coronary ischemia

Angina pectoris

Atherosclerosis
and exertion

Typical angina

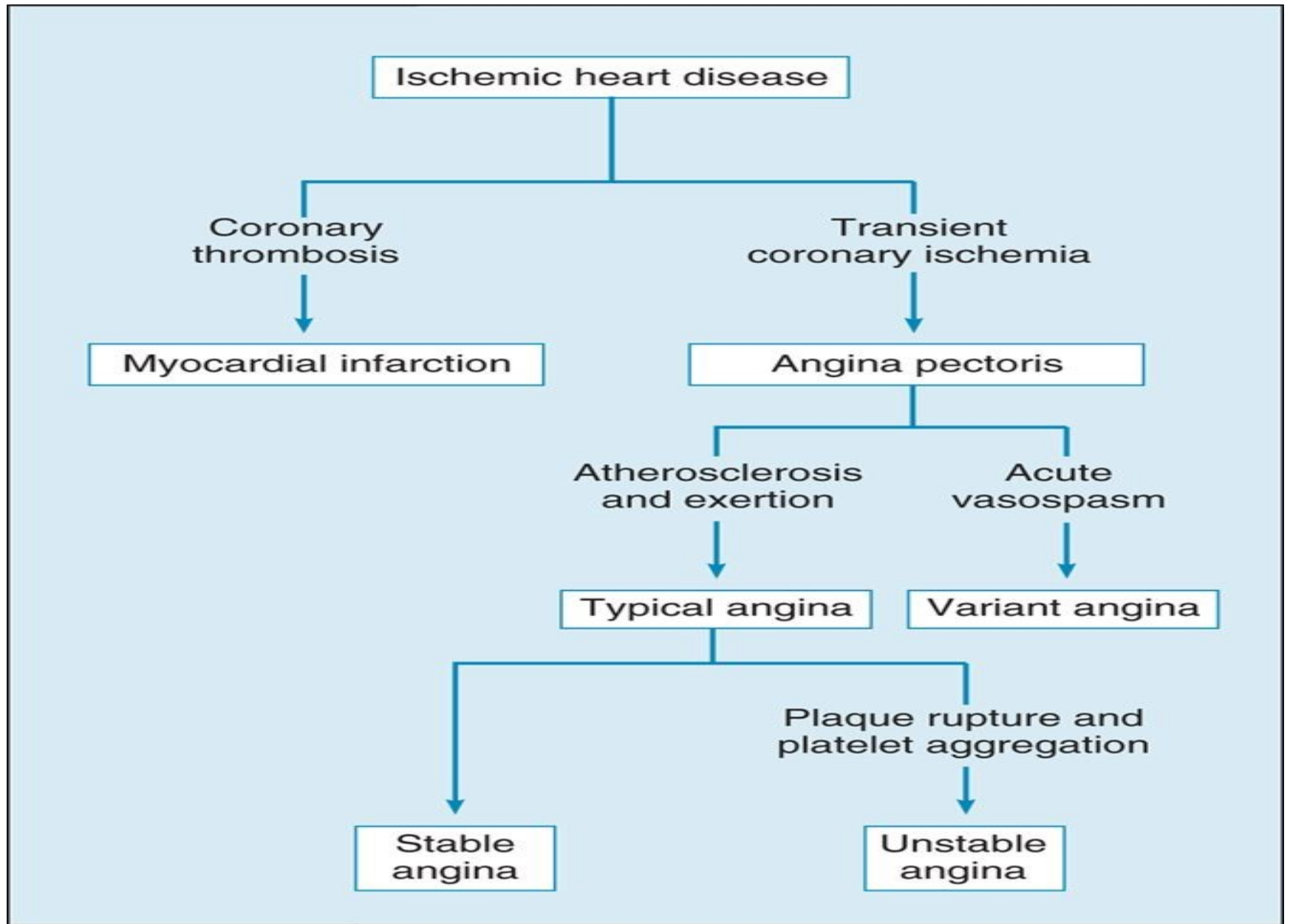
Acute
vasospasm

Variant angina

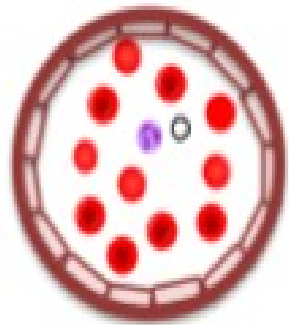
Plaque rupture and
platelet aggregation

Stable
angina

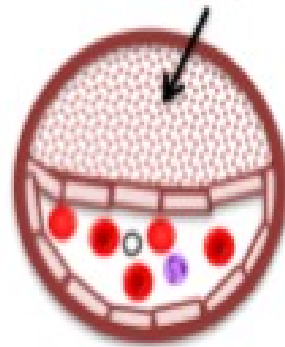
Unstable
angina



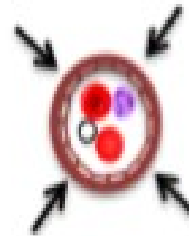
Atherosclerotic
plaque



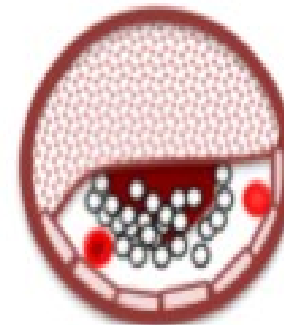
Normal



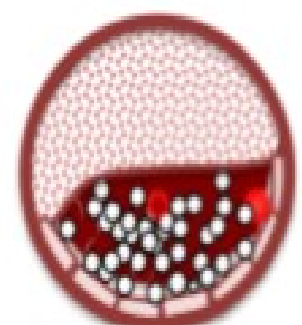
Exertional
Angina



Vasospastic
Angina



Unstable
Angina



Infarct



Thrombus



Endothelial cell



Neutrophil



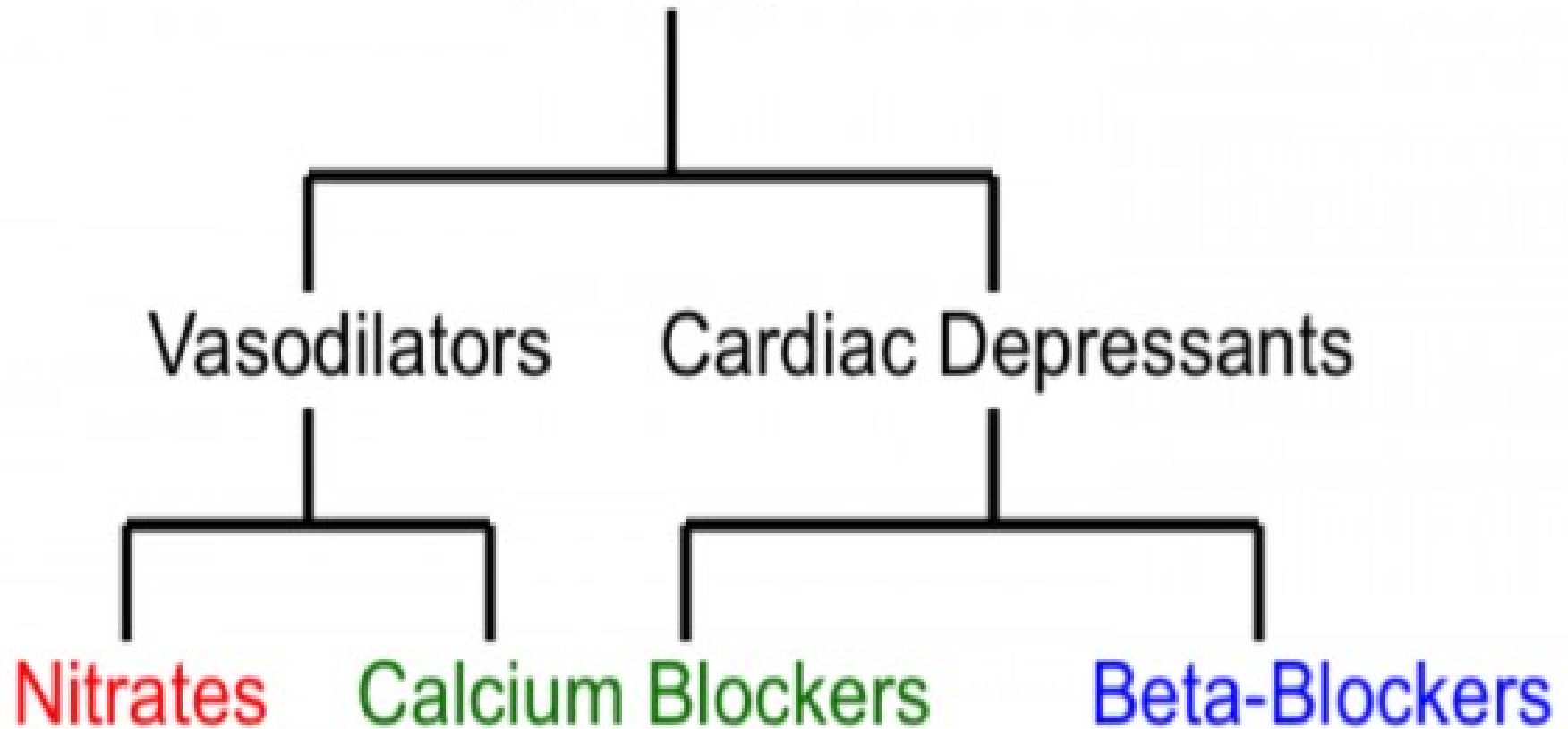
Platelet



RBC

- **Two goals of angina drug therapy**
 - Prevention of myocardial infarction and death
 - Prevention of myocardial ischemia and anginal pain
- **Three families of anti-anginal agents**
 - Organic nitrates
 - Beta blockers
 - Calcium channel blockers

Most Common Drugs Used in Treating Angina Pectoris



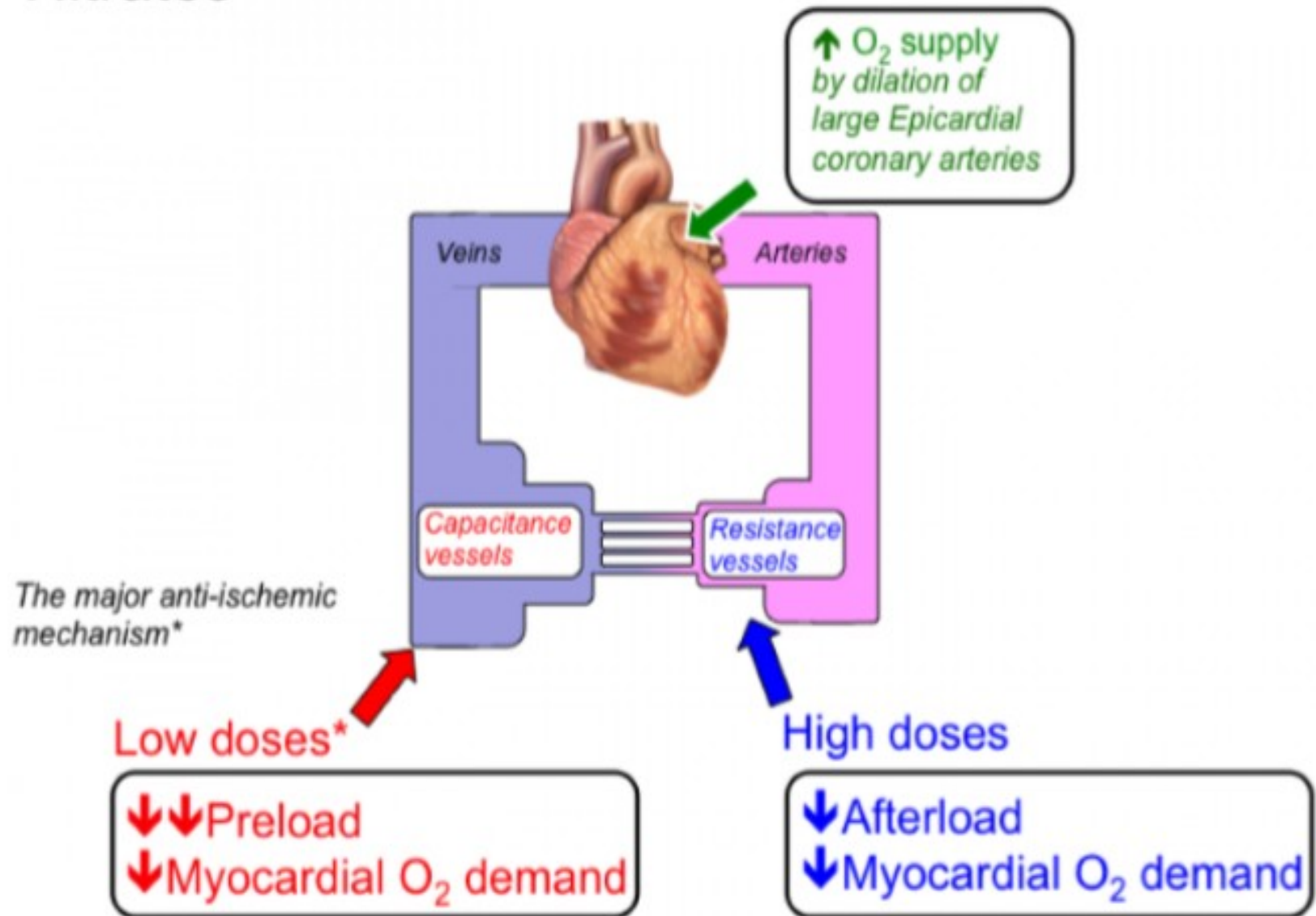
I. Organic nitrates

- the oldest and most frequently used antianginal drugs
- Work by causing vasodilation
- drugs
 - nitroglycerin
 - isosorbide dinitrate
 - isosorbide mononitrate
 - amyl nitrite

Mechanism of action

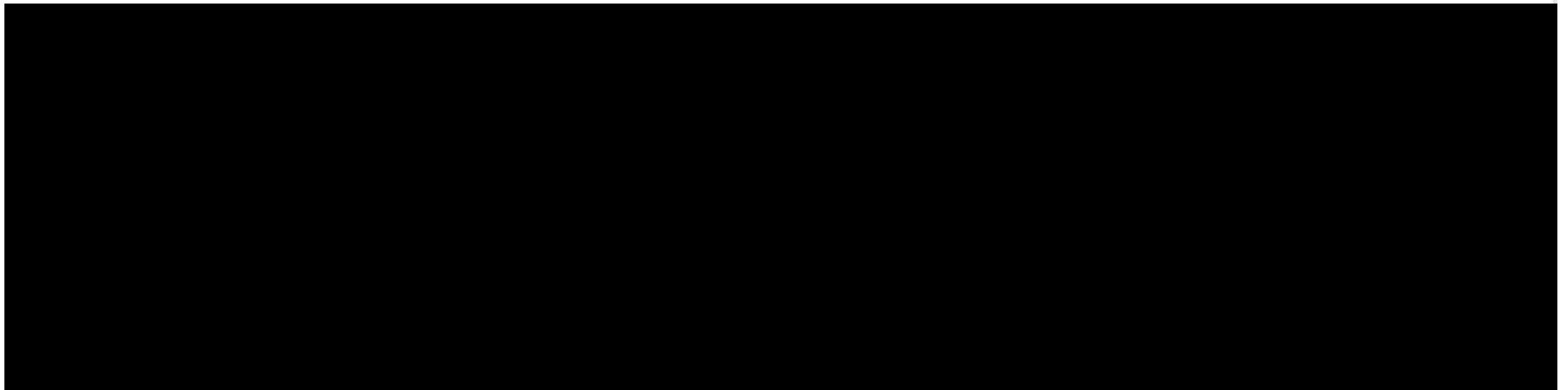
- inhibit coronary vasoconstriction or spasm, increasing perfusion of the myocardium and,
 - thus, relieving **vasospastic angina**
- relax the veins (venodilation), decreasing preload and myocardial oxygen consumption
 - are effective in treating effort-induced angina (**classic angina**)

Nitrates

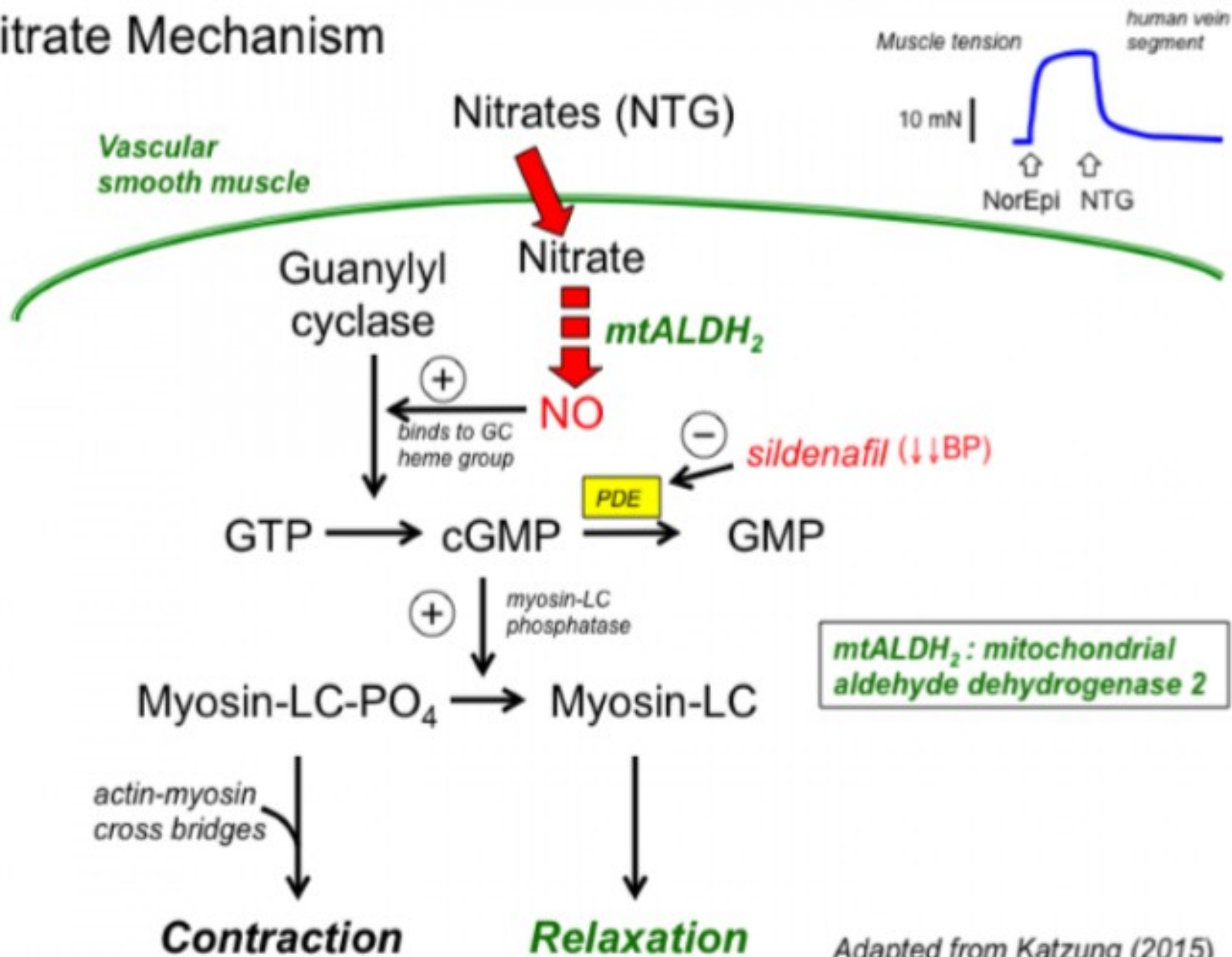


MOA

- are enzymatically broken down in the vascular wall by both glutathione S-transferase (GST) and mitochondrial aldehyde dehydrogenase isoform 2 (MtADH2) to release the vasodilator substance “nitric oxide”



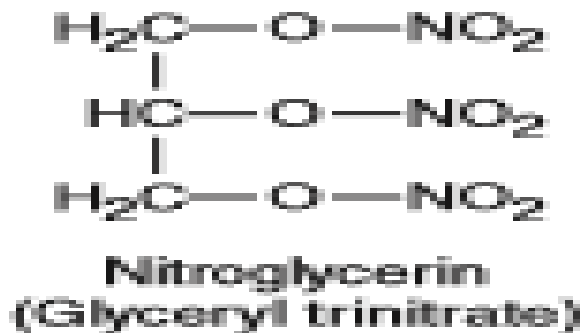
Nitrate Mechanism



Adapted from Katzung (2015)

Pharmacokinetics

- The time to onset of action varies from **1 minute for nitroglycerin** to more than 1 hour for isosorbide mononitrate
- **Significant first-pass metabolism of nitroglycerin occurs in the liver**
 - Therefore, it is common to take the drug either sublingually or via a transdermal patch, thereby avoiding this route of elimination



Adverse effects

- Most common-headache
- Dizziness
- postural hypotension
- facial flushing
- Reflex tachycardia

- Combination with Phosphodiesterase V inhibitors such as sildenafil, tadalafil, and vardenafil is contraindicated
 - **dangerous hypotension**
 - nitroglycerin should not be used within 24 hours after taking sildenafil (Viagra[®]) or within 2 days after taking tadalafil

Tolerance

- Tolerance to the actions of nitrates develops rapidly
- The blood vessels become desensitized to vasodilation
- can be overcome by providing a daily “**nitrate-free interval**” to restore sensitivity to the drug
 - **10 to 12 hours, usually at night**

TABLE 12-2 Nitrate and nitrite drugs used in the treatment of angina

Abbreviations: IV, intravenous; PO, per os; sublingual, sublingual; SL, sublingual; subcutaneous, subcutaneous; subcutaneous, subcutaneous; subcutaneous, subcutaneous.

Drug	Formulation	Dose	Onset of Action	Duration of Action	Contraindications	Side Effects
Nitroglycerin	Sublingual tablet, sublingual spray, transdermal patch, intravenous solution	Sublingual: 0.25 to 0.5 mg, repeat every 5 minutes as needed. Transdermal: 0.2 to 0.8 mg/hr. Intravenous: 0.1 to 0.4 mg/min.	Sublingual: 1 to 3 minutes. Transdermal: 30 to 60 minutes. Intravenous: 1 to 3 minutes.	Sublingual: 3 to 5 minutes. Transdermal: 12 to 24 hours. Intravenous: 1 to 3 minutes.	Hypotension, headache, dizziness, flushing, tachycardia, tolerance.	Hypotension, headache, dizziness, flushing, tachycardia, tolerance.
Isosorbide dinitrate	Sublingual tablet, transdermal patch, intravenous solution	Sublingual: 5 to 10 mg, repeat every 5 minutes as needed. Transdermal: 2 to 8 mg/hr. Intravenous: 0.1 to 0.4 mg/min.	Sublingual: 1 to 3 minutes. Transdermal: 30 to 60 minutes. Intravenous: 1 to 3 minutes.	Sublingual: 3 to 5 minutes. Transdermal: 12 to 24 hours. Intravenous: 1 to 3 minutes.	Hypotension, headache, dizziness, flushing, tachycardia, tolerance.	Hypotension, headache, dizziness, flushing, tachycardia, tolerance.
Isosorbide mononitrate	Sublingual tablet, transdermal patch, intravenous solution	Sublingual: 5 to 10 mg, repeat every 5 minutes as needed. Transdermal: 2 to 8 mg/hr. Intravenous: 0.1 to 0.4 mg/min.	Sublingual: 1 to 3 minutes. Transdermal: 30 to 60 minutes. Intravenous: 1 to 3 minutes.	Sublingual: 3 to 5 minutes. Transdermal: 12 to 24 hours. Intravenous: 1 to 3 minutes.	Hypotension, headache, dizziness, flushing, tachycardia, tolerance.	Hypotension, headache, dizziness, flushing, tachycardia, tolerance.
Sodium nitroprusside	Intravenous solution	0.1 to 0.4 mg/min.	1 to 3 minutes.	1 to 3 minutes.	Hypotension, headache, dizziness, flushing, tachycardia, tolerance.	Hypotension, headache, dizziness, flushing, tachycardia, tolerance.

II. Beta blockers

- decrease the oxygen demands of the myocardium by lowering both the rate and the force of contraction of the heart
- drugs of choice to treat effort-induced angina
- ineffective against and should not be used in vasospastic angina
- Approved for Angina are atenolol, metoprolol, propranolol
- Cardio selective β -blockers, such as *metoprolol and atenolol*, are preferred
- CI in **asthma, diabetes, severe bradycardia, and chronic obstructive pulmonary disease**

- Adding a beta blocker can reduce the need for nitroglycerin PRN by **decreasing myocardial work and oxygen demand** (by reducing heart rate, contractility, and afterload) that contribute to producing angina

III. Calcium channel blockers

- **Arteriolar vasodilators:** decrease in smooth muscle tone and vascular resistance
- **Decrease after load-effective in effort-induced angina**
- Relaxation of the coronary arteries-**vasospastic angina**
- verapamil, diltiazem, and nifedipine, amlodipine
- **Verapamil and diltiazem** can produce modest additional reductions in oxygen demand by suppressing heart rate and contractility

Calcium Channel Blocker Side Effects:

- Constipation
- flushing & dizziness
- AV & SA node depression (more common with verapamil)
- **dihydropyridines** can cause **reflex sympathetic stimulation of the heart** due to their vascular selective effects
 - combining them with a beta blocker can overcome this unwanted side effect.

Oral

Half-Life

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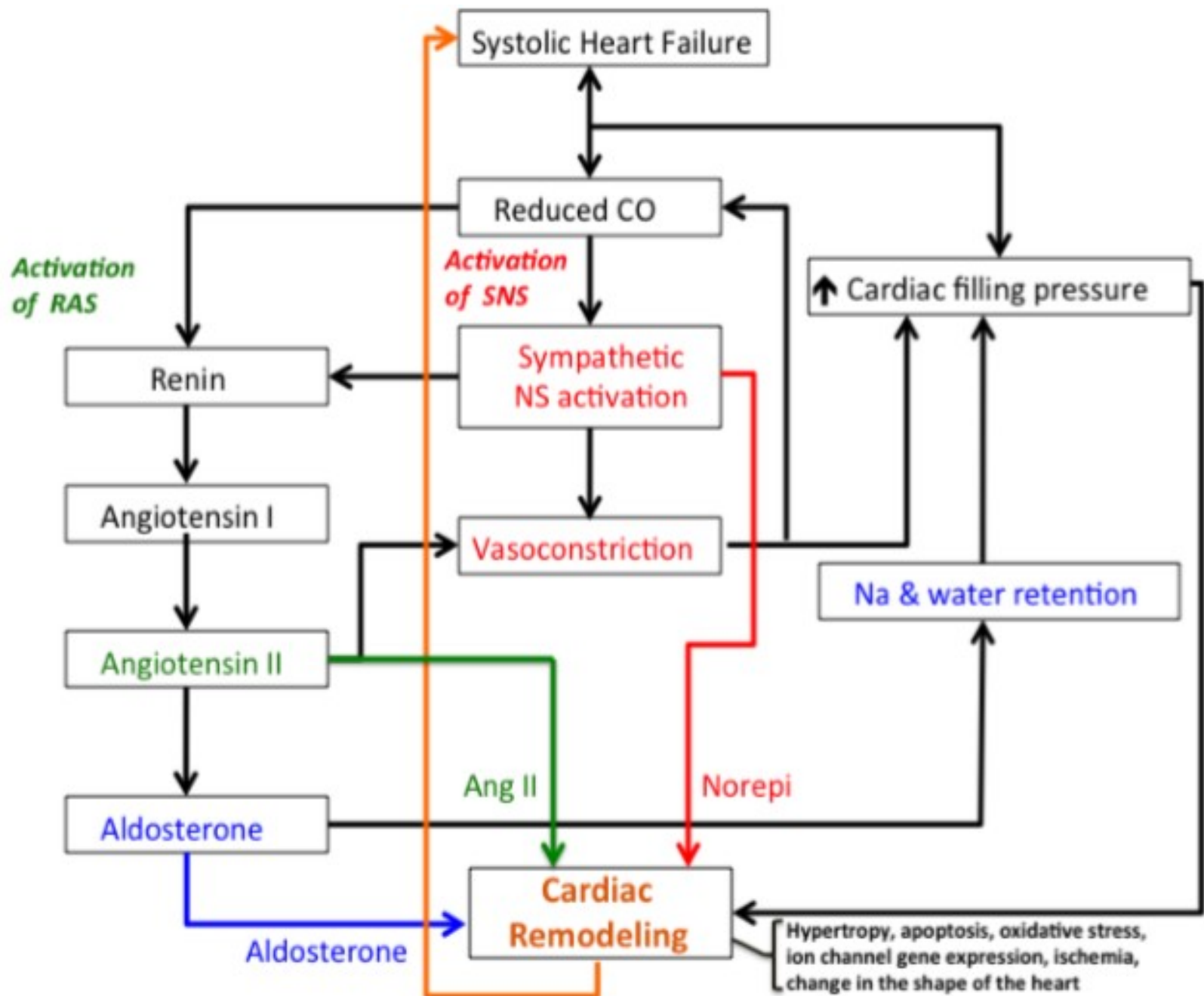
Drugs for heart failure

What is HF?

- a syndrome in which the heart is **unable to pump sufficient blood to meet the metabolic needs of tissues**
- It is a highly lethal condition,
 - with a 5-year mortality rate conventionally said to be about 50%
- most common causes
 - **coronary artery disease**
 - hypertension

signs and symptoms

- decreased exercise tolerance with rapid muscular fatigue
 - direct consequence of **diminished cardiac output**
- Tachycardia
- shortness of breath
- peripheral and pulmonary edema, and cardiomegaly
 - Resulting from **compensatory responses to reduced output**



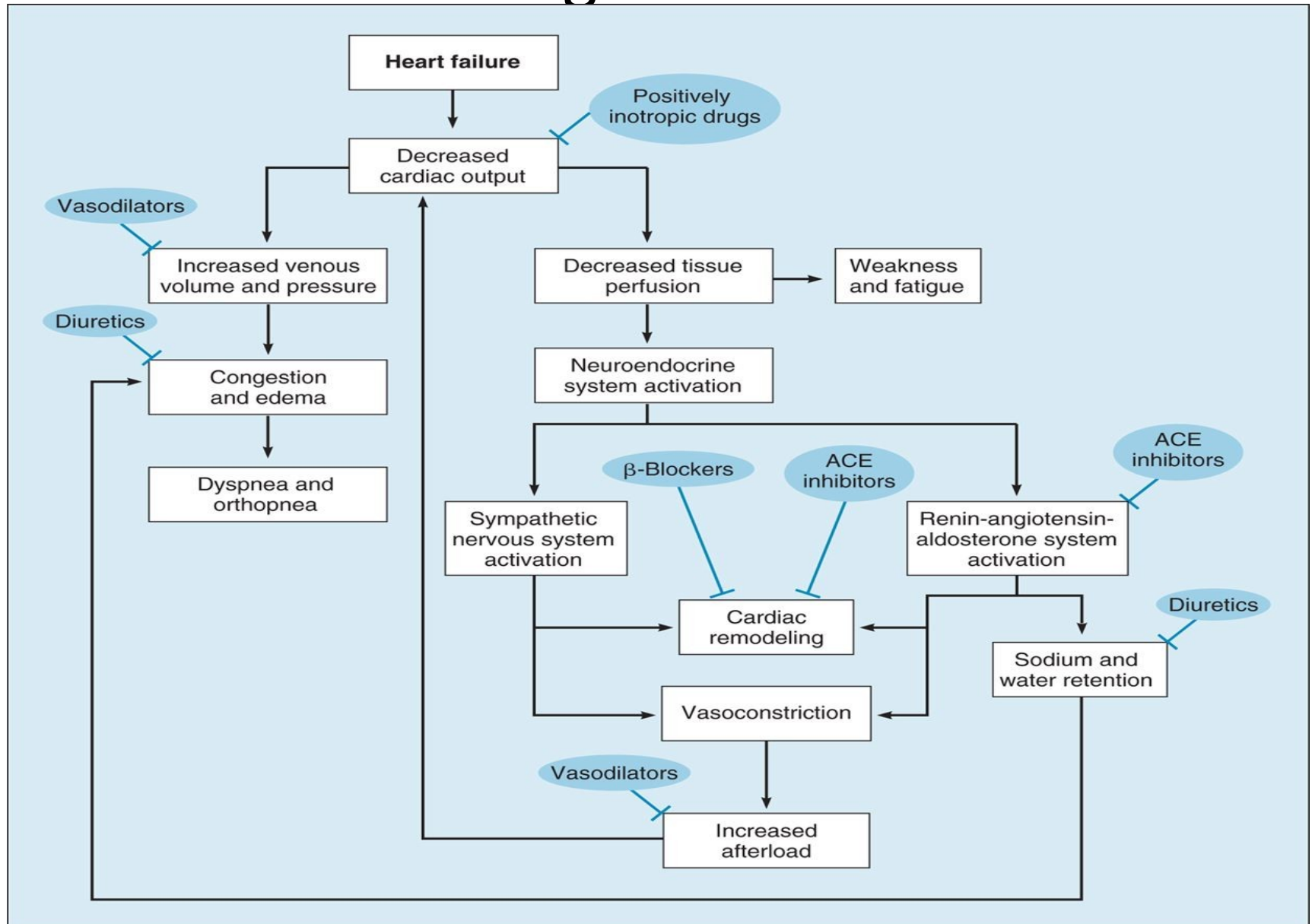
- **Remodeling**

- is the term applied to dilation (other than that due to passive stretch) and
- other slow structural changes that occur in the stressed myocardium

Drugs for heart failure

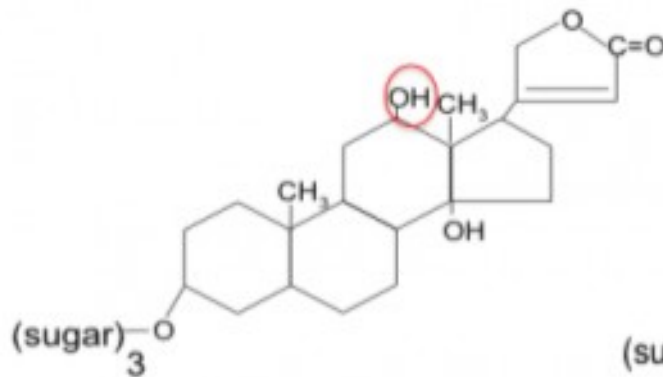
- **Positive Inotropics**
 - Digoxin and other cardiac glycosides
 - Other Inotropic agents
- **Not Positive Inotropics**
 - Diuretics
 - ACE inhibitors
 - angiotensin receptor antagonists
 - aldosterone antagonists, and
 - β blockers

Drugs for HF

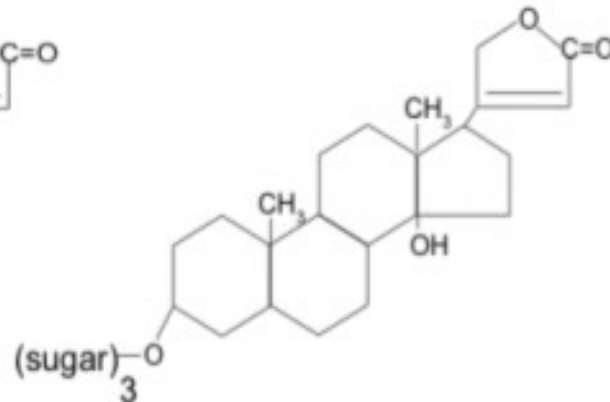


Cardiac glycosides

- chemically related compounds isolated primarily from plant sources
 - purple and white foxglove plants (*Digitalis purpurea* and *Digitalis lanata*)



Digoxin



Digitoxin

Pharmacokinetics

Table 1: Pharmacokinetic Properties of Digitalis Glycosides

Property	Digoxin	Digitoxin
Lipid Solubility	medium	high
Oral Absorption	75%	>90%
Plasma Protein Binding	20-40%	>90%
Plasma Half-Life	1.6 days	7 days
Route of Elimination	Kidney	Liver

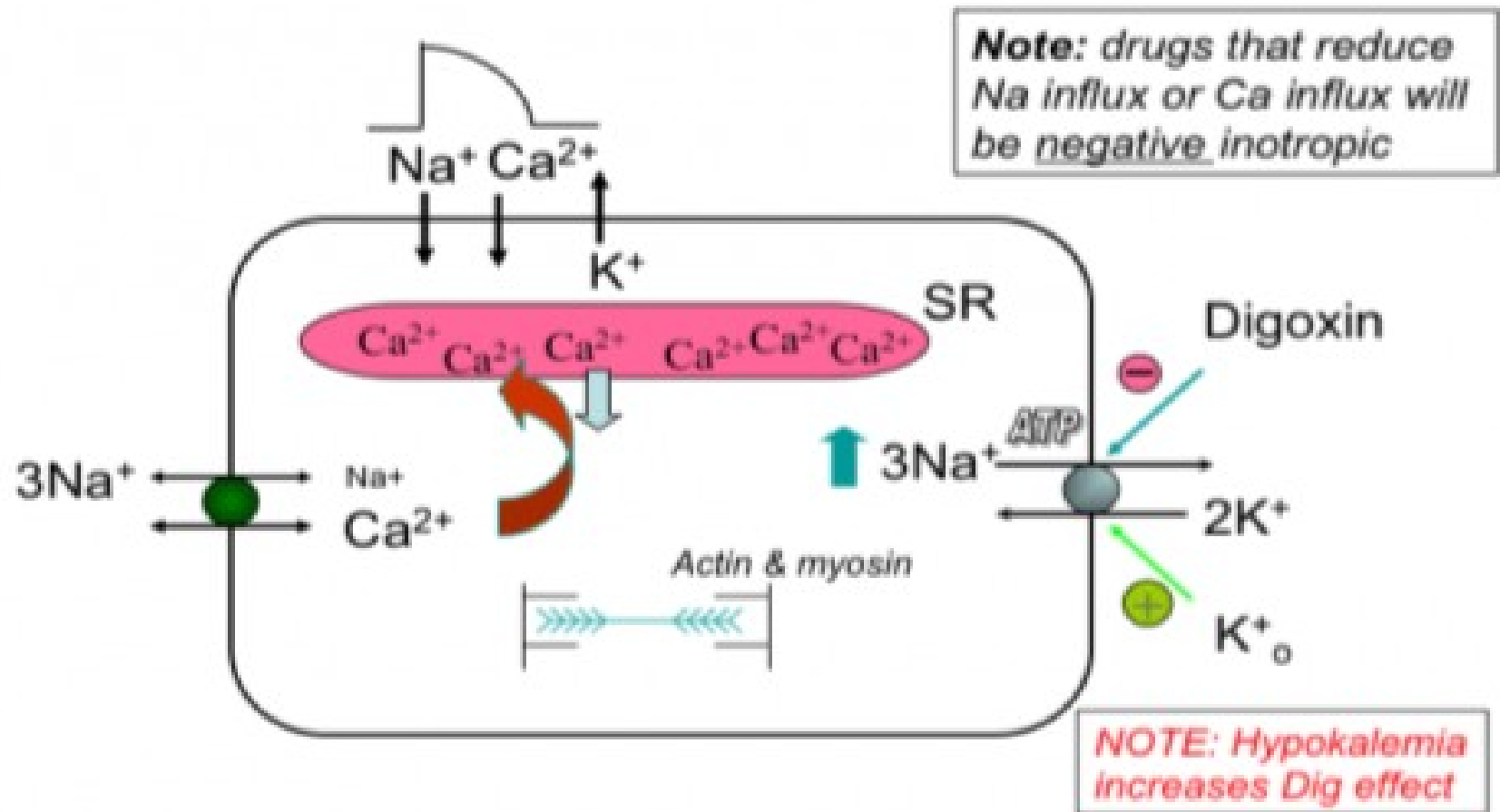
Pharmacodynamics

- At therapeutic doses, **digoxin increases vagal tone** to the heart and thereby **increasing the ERP of the AVN**
 - mechanisms
 - sensitization of baroreceptors
 - central vagal stimulation
 - facilitation of muscarinic transmission to the heart
 - benefit
 - reducing and controlling the ventricular rate in patients suffering from **atrial fibrillation or flutter**, or **abolishing AVN reentry**.

- Digoxin and the other cardiac glycosides have positive inotropic actions
 - i.e. increase myocardial contractile force
 - Thereby increase cardiac output
 - significant diuresis following the improvement in kidney perfusion
- can reduce symptoms of HF, but they do not prolong life

MOA

Mechanism of Positive Inotropic Action



- Digoxin is given PO or IV
- Digitalization is accomplished rapidly by giving a total dose of 0.75-1.5 mg in divided doses 6-8 hrs apart over 24 hrs.
- The target plasma drug level is 0.5 to 0.8 ng/mL.
- Drug elimination takes approximately 1 week.

Digoxin toxicity

Cardiac Toxicity

2nd or 3rd degree AV conduction block, Bradycardia (caused by increased vagal tone)

Ventricular arrhythmias (extrasystoles, bigeminy, VTach) (likely caused by DADs)

ECG changes (↑PR, ST depression, inverted T wave, ↓QT, P wave changes)

Non-Cardiac Signs/Symptoms of Toxicity

Nausea/vomiting

Anorexia

Fatigue

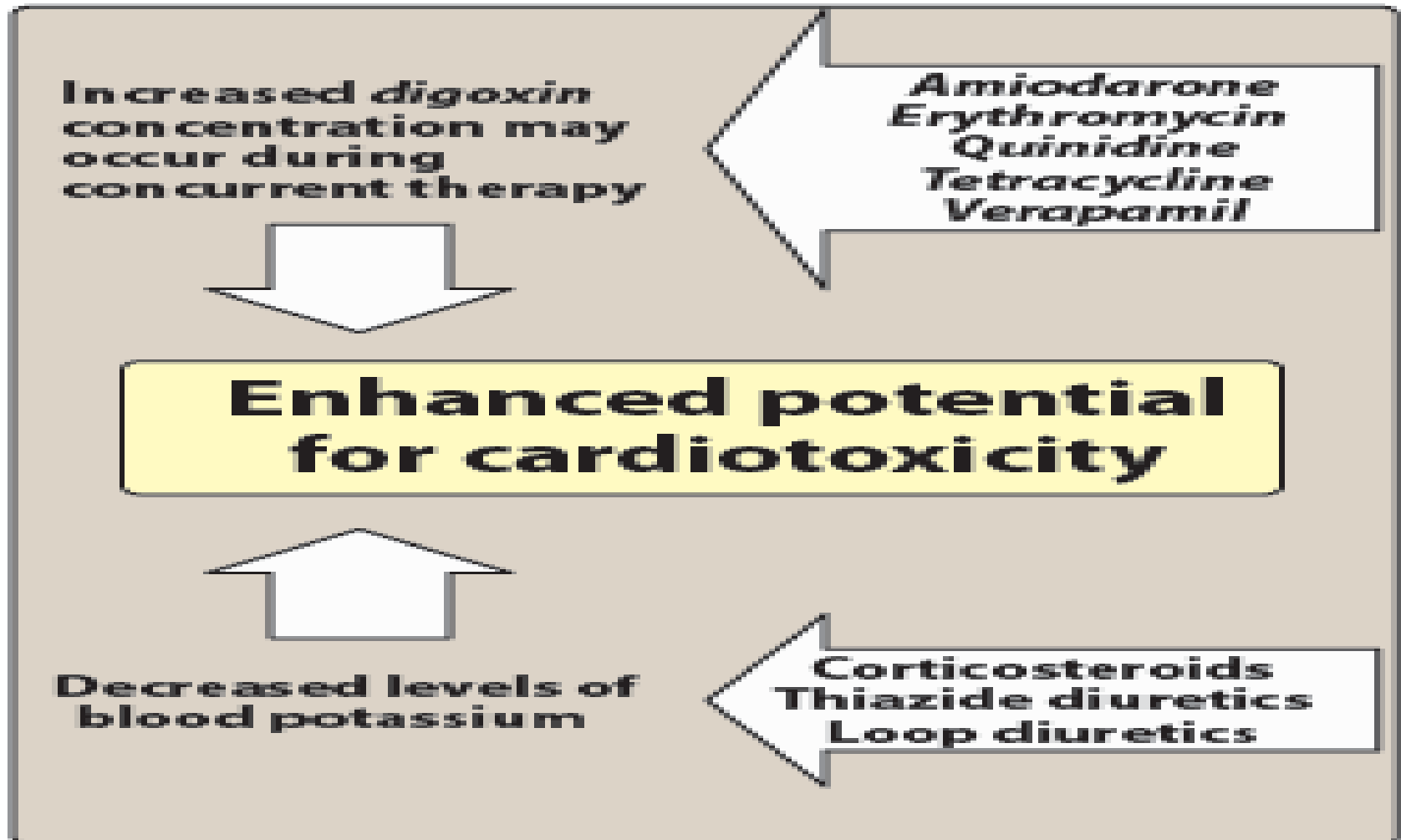
Visual complaints (yellow halo)

- Severe toxicity resulting in ventricular tachycardia may require administration of
 - antiarrhythmic drugs
 - lidocaine or phenytoin (for ventricular arrhythmias)
 - atropine (for AVN block)
 - the use of antibodies to digoxin (**digoxin immune Fab**), which bind and inactivate the drug

Drug interaction

- **decreased K⁺** increases the potential for cardiotoxicity
- Reduction of serum potassium levels is most frequently observed in patients receiving **thiazide or loop diuretics**
 - which can usually be prevented by use of a potassium-sparing diuretic or supplementation with potassium chloride
- Hypocalcemia decreases the effect/toxicity of digoxin

Drugs interacting with digoxin



Other inotropic drugs

- In addition to the cardiac glycosides, there are other groups of inotropic drugs
 - phosphodiesterase (PDE 3) inhibitors
- **milrinone and Inamrinone** (has been withdrawn from the market)
- Milrinone less likely to cause bone marrow and liver toxicity, but it does cause arrhythmias.
- Milrinone is now used only intravenously and only for acute heart failure or severe exacerbation of chronic heart failure.

Phosphodiesterase Inhibitors

- increase myocardial contractility
 - By causing intracellular accumulation of cyclic AMP (cAMP) secondary to inhibition of PDE3, an enzyme that degrades cAMP
- long-term therapy may be associated with a substantial increase in the risk of mortality

Sympathomimetic Drugs

- Dopamine and Dobutamine
- Dopamine activate
 - beta1-adrenergic receptors in the heart,
 - dopamine receptors in the kidney, and
 - at high doses, alpha1-adrenergic receptors in blood vessels
- Dobutamine causes selective activation of beta1-adrenergic receptors

- Dobutamine produces angina or arrhythmias in patients with coronary artery disease
- Intermittent dobutamine infusion may benefit some patients with chronic heart failure.

Diuretics

- are the mainstay of heart failure management
- **The thiazide diuretics** (eg, hydrochlorothiazide)
 - produce moderate diuresis
 - used for long-term therapy of HF when edema is not too great
- **The loop diuretics** (eg, furosemide)
 - produce profound diuresis
 - In contrast to the thiazides, these drugs can promote fluid loss even when GFR is low
 - **drugs of choice for patients with severe HF**

- can cause **hypokalemia**, thereby increasing the risk of digoxin toxicity
- loop diuretics can also cause severe **hypotension** secondary to excessive volume reduction
- **Potassium-Sparing diuretics**
 - promote only scant diuresis
 - employed to counteract potassium loss caused by thiazide and loop diuretics
 - **Cause hyperkalemia**
 - Shouldn't be combined with ACEIs and ARBs

- **spironolactone**—prolongs survival in patients with HF primarily by blocking receptors for aldosterone

ACEIs

- they improve hemodynamics and favorably alter cardiac remodeling
- reduce peripheral resistance
 - thereby reduce afterload
- reduce salt and water retention (by reducing aldosterone secretion)
 - there by reduce preload
- reduce the long-term remodeling of the heart and vessels
 - an effect that may be responsible for the observed reduction in **mortality and morbidity**

Angiotensin II Receptor Blockers

- the effects of ARBs are similar to those of ACE inhibitors—but not identical
- because ARBs do not increase levels of kinins, their effects on cardiac remodeling are less favorable than those of ACE inhibitors
- should be considered in patients intolerant of ACE inhibitors because of incessant cough

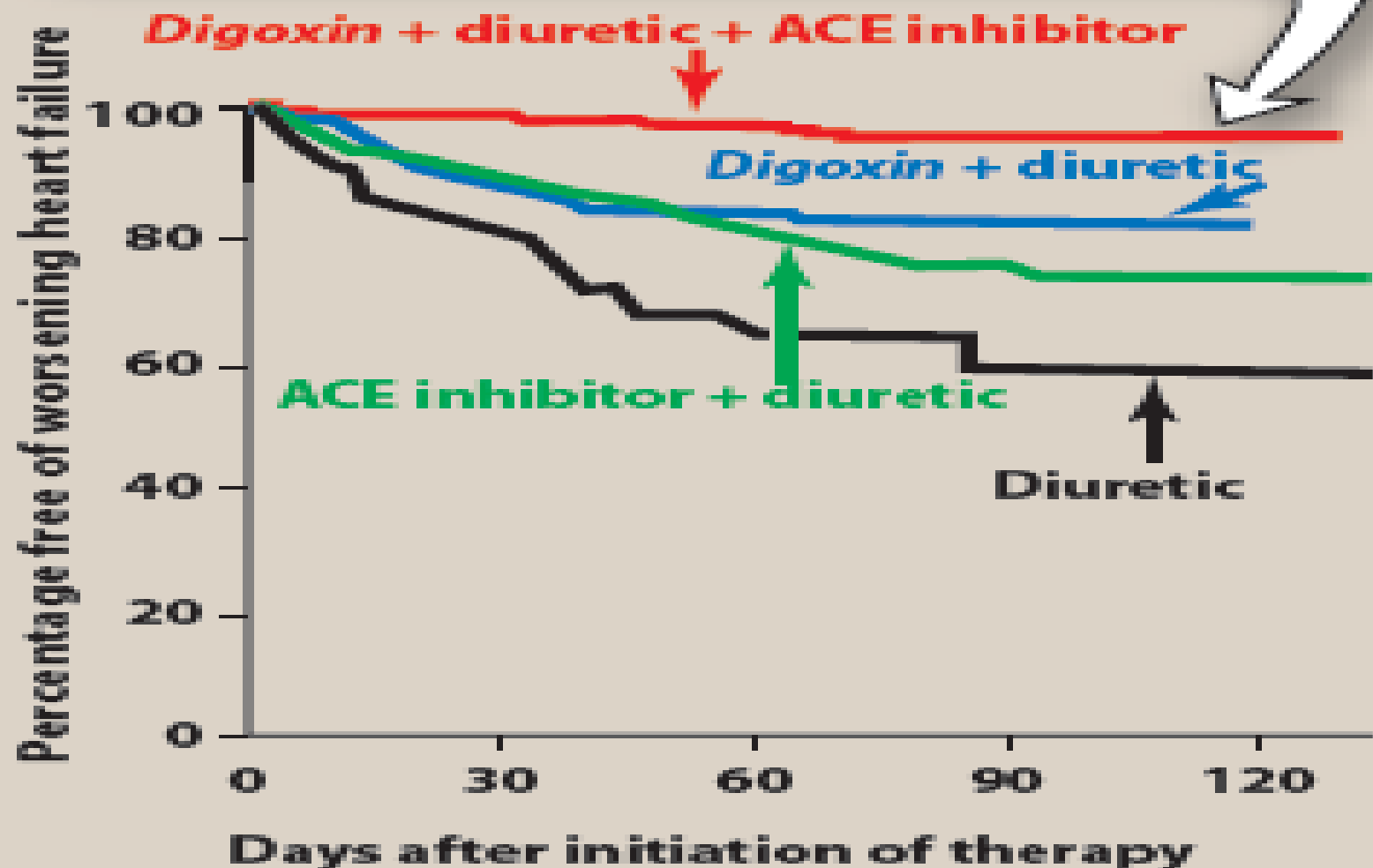
Beta-Adrenoceptor Blockers

- In the past HF was considered an absolute contraindication to these drugs
- **bisoprolol, carvedilol, and metoprolol** showed a reduction in mortality in patients with stable severe heart failure
- the mechanism underlying benefits is uncertain
- **suggested mechanisms include**
 - attenuation of the adverse effects of high concentrations of catecholamines (including apoptosis)
 - up-regulation of receptors
 - decreased heart rate and
 - reduced remodeling through inhibition of the mitogenic activity of catecholamines

Other Vasodilators

- are effective in acute heart failure because they provide
 - a reduction in preload (through venodilation), or
 - reduction in afterload (through arteriolar dilation), or both
- If the patient is intolerant of ACE inhibitors or β -blockers, or if additional vasodilator response is required,
 - a combination of hydralazine and isosorbide dinitrate may be used

Patients receiving triple therapy were least likely to experience worsening of heart failure compared to any of the other three treatment groups.



Thank u!